

Pandas Interview Practice Guide

CSV Datasets and Comprehensive Questions for Data Analyst Interviews

Dataset 1: Sales Data (sales_data.csv)

CSV

employee_id	employee_name	department	region	sale_date	product_category	product_name	quantity	unit_price	total_price	discount	commission
101	John Smith	Sales	North	2024-01-15	Electronics	Laptop	2	999.99	1999.98	C001	
102	Sarah Johnson	Sales	South	2024-01-16	Electronics	Phone	1	599.99	599.99	C002	
103	Mike Brown	Sales	East	2024-01-17	Clothing	Shirt	3	29.99	89.97	C003	
101	John Smith	Sales	North	2024-01-18	Electronics	Tablet	1	299.99	299.99	C004	
104	Lisa Davis	Sales	West	2024-01-19	Home	Chair	2	149.99	299.98	C005	
102	Sarah Johnson	Sales	South	2024-01-20	Electronics	Laptop	1	999.99	999.99	C006	
105	Tom Wilson	Sales	North	2024-01-21	Clothing	Pants	2	49.99	99.98	C007	
103	Mike Brown	Sales	East	2024-01-22	Home	Table	1	399.99	399.99	C008	
106	Amy Taylor	Sales	West	2024-01-23	Electronics	Phone	2	599.99	1199.98	C009	
104	Lisa Davis	Sales	West	2024-01-24	Clothing	Dress	1	79.99	79.99	C010	
101	John Smith	Sales	North	2024-02-01	Electronics	Laptop	1	999.99	999.99	C011	
107	David Lee	Sales	South	2024-02-02	Home	Sofa	1	799.99	799.99	C012	
102	Sarah Johnson	Sales	South	2024-02-03	Electronics	Tablet	3	299.99	899.97	C013	
108	Emma White	Sales	East	2024-02-04	Clothing	Jacket	1	129.99	129.99	C014	
105	Tom Wilson	Sales	North	2024-02-05	Electronics	Phone	1	599.99	599.99	C015	

Dataset 2: Employee Performance (employee_performance.csv)

CSV

employee_id	employee_name	department	hire_date	salary	performance_score	training_hours	projects_completed
101	John Smith	Sales	2022-03-15	65000	4.2	40	12
102	Sarah Johnson	Sales	2021-07-20	58000	4.5	35	15
103	Mike Brown	Sales	2023-01-10	52000	3.8	25	8
104	Lisa Davis	Sales	2022-11-05	60000	4.1	30	10
105	Tom Wilson	Sales	2021-12-12	63000	4.3	45	14
106	Amy Taylor	Sales	2023-05-18	55000	3.9	20	7
107	David Lee	Sales	2022-09-30	59000	4.0	38	11
108	Emma White	Sales	2023-02-14	53000	4.4	42	9
201	Robert Garcia	Management	2019-01-15	85000	4.6	60	20
202	Jennifer Kim	Management	2020-03-20	82000	4.4	55	18
301	Michael Johnson	Executive	2018-06-10	120000	4.8	70	25

Dataset 3: Customer Data (customer_data.csv)

CSV

```
customer_id,customer_name,age,gender,city,state,registration_date,last_purchase_date,total_purc
C001,Alice Brown,28,Female,New York,NY,2023-06-15,2024-01-15,5,2500.50
C002,Bob Wilson,35,Male,Los Angeles,CA,2023-03-20,2024-01-16,3,1800.75
C003,Carol Davis,42,Female,Chicago,IL,2023-08-10,2024-01-17,7,3200.25
C004,Daniel Lee,31,Male,Houston,TX,2023-05-05,2024-01-18,4,1950.00
C005,Eva Martinez,26,Female,Phoenix,AZ,2023-09-12,2024-01-19,6,2750.80
C006,Frank Taylor,39,Male,Philadelphia,PA,2023-04-18,2024-01-20,8,4100.60
C007,Grace Chen,33,Female,San Antonio,TX,2023-07-22,2024-01-21,5,2300.40
C008,Henry Johnson,45,Male,San Diego,CA,2023-02-28,2024-01-22,9,4500.25
C009,Ivy Thompson,29,Female,Dallas,TX,2023-10-15,2024-01-23,3,1650.75
C010,Jack Robinson,37,Male,San Jose,CA,2023-01-30,2024-01-24,6,3150.90
C011,Kate Anderson,24,Female,Austin,TX,2023-11-08,2024-02-01,2,1200.30
C012,Liam Garcia,41,Male,Jacksonville,FL,2023-06-25,2024-02-02,7,3800.50
C013,Mia Rodriguez,32,Female,Fort Worth,TX,2023-08-14,2024-02-03,4,2100.85
C014,Noah Martinez,36,Male,Columbus,OH,2023-03-07,2024-02-04,5,2650.40
C015,Olivia Wilson,30,Female,Charlotte,NC,2023-09-19,2024-02-05,6,2980.70
```



Dataset 4: Product Inventory (product_inventory.csv)

CSV

```
product_id,product_name,category,brand,cost_price,selling_price,stock_quantity,reorder_level,su
P001,Laptop,Electronics,TechBrand,750.00,999.99,45,10,S001,2024-01-10
P002,Phone,Electronics,PhoneCorp,400.00,599.99,32,15,S002,2024-01-12
P003,Tablet,Electronics,TechBrand,200.00,299.99,28,8,S001,2024-01-14
P004,Shirt,Clothing,FashionCo,15.00,29.99,85,20,S003,2024-01-08
P005,Pants,Clothing,StyleBrand,25.00,49.99,62,15,S004,2024-01-09
P006,Dress,Clothing,FashionCo,40.00,79.99,23,10,S003,2024-01-11
P007,Jacket,Clothing,StyleBrand,70.00,129.99,18,8,S004,2024-01-13
P008,Chair,Home,HomePlus,80.00,149.99,15,5,S005,2024-01-07
P009,Table,Home,FurnitureCo,250.00,399.99,8,3,S006,2024-01-06
P010,Sofa,Home,HomePlus,500.00,799.99,4,2,S005,2024-01-05
```



COMPREHENSIVE PRACTICE QUESTIONS

SECTION 1: Data Loading and Basic Operations

- 1. Load all CSV files into pandas DataFrames and display basic information about each dataset.
- 2. Check the shape, data types, and memory usage of each DataFrame.
- 3. Identify and handle missing values in all datasets. Create a summary report.
- 4. Display the first 5 and last 3 rows of the sales_data DataFrame.

5. Get basic statistical summary for numerical columns in employee_performance data.

SECTION 2: Data Filtering and Selection

6. Find all sales records for Electronics category with total_amount > 500.
7. Select employees hired after 2022 with performance_score >= 4.0.
8. Filter customers who are female, aged between 25-35, and from Texas.
9. Find products with stock_quantity below reorder_level.
10. Get sales data for specific employees: John Smith, Sarah Johnson, and Mike Brown.

SECTION 3: Data Aggregation and Grouping

11. Calculate total sales amount by region and product category.
12. Find average salary and performance score by department.
13. Count number of customers by state and gender.
14. Calculate monthly sales totals for January and February 2024.
15. Find top 3 performing employees by total sales amount.

SECTION 4: Data Merging and Joining

16. Merge sales_data with employee_performance to get employee details with their sales.
17. Join sales_data with customer_data to analyze customer purchasing patterns.
18. Create a comprehensive dataset by merging sales with both employee and customer data.
19. Left join employee_performance with sales_data to find employees with no sales.
20. Merge product_inventory with sales_data to analyze product performance.

SECTION 5: Date and Time Operations

21. Convert date columns to datetime format in all relevant datasets.
22. Extract year, month, and day from sale_date in sales_data.
23. Calculate days between hire_date and current date for all employees.
24. Find customers who made their last purchase more than 30 days ago.
25. Group sales by week and calculate weekly totals.

SECTION 6: Advanced Data Manipulation

26. Create age groups for customers: Young (18-30), Middle (31-45), Senior (46+).
27. Calculate profit margin for each product (selling_price - cost_price).
28. Rank employees by total sales amount within each region.
29. Create a new column showing percentage of total sales for each transaction.

30. Calculate running total of sales for each employee.

SECTION 7: Statistical Analysis

31. Find correlation between employee salary and performance_score.

32. Calculate quartiles for customer_lifetime_value.

33. Identify outliers in sales amounts using IQR method.

34. Perform statistical tests on sales performance by region.

35. Calculate standard deviation of sales by product category.

SECTION 8: Data Transformation and Reshaping

36. Pivot sales data to show total sales by employee and product category.

37. Melt employee_performance data to create a long format for analysis.

38. Create a cross-tabulation of customer gender vs state.

39. Stack and unstack operations on sales data by region and category.

40. Transform wide customer data to long format for visualization.

SECTION 9: Complex Business Questions

41. Identify the most profitable product category by region.

42. Find employees who consistently perform above average across multiple metrics.

43. Calculate customer retention rate by analyzing purchase patterns.

44. Identify products that need urgent restocking based on sales velocity.

45. Find the best performing sales team (by manager) in terms of revenue and customer satisfaction.

SECTION 10: Data Quality and Validation

46. Check for duplicate records across all datasets and handle them appropriately.

47. Validate data integrity between related tables (referential integrity).

48. Identify and handle inconsistent data formats across datasets.

49. Create data quality reports showing completeness and accuracy metrics.

50. Implement data validation rules and flag violations.

SECTION 11: Performance Optimization

51. Optimize DataFrame operations for large datasets using vectorization.

52. Use categorical data types where appropriate to reduce memory usage.

53. Implement efficient filtering strategies for large datasets.

54. Compare performance of different aggregation methods.

55. **Optimize join operations for better performance.**

SECTION 12: Advanced Analytics

56. **Calculate year-over-year growth rates for sales.**

57. **Implement customer segmentation using RFM analysis.**

58. **Create cohort analysis for customer retention.**

59. **Perform ABC analysis on products based on sales volume.**

60. **Calculate employee productivity metrics and efficiency ratios.**

SAMPLE SOLUTIONS FOR KEY QUESTIONS

Question 11 Solution: Total sales by region and category

python

```
import pandas as pd
```

```
# Load data
```

```
sales_df = pd.read_csv('sales_data.csv')
```

```
# Calculate total sales by region and product category
```

```
sales_summary = sales_df.groupby(['region', 'product_category'])['total_amount'].sum().reset_index()
```

```
sales_pivot = sales_summary.pivot(index='region', columns='product_category', values='total_amount')
```

```
print(sales_pivot)
```

Question 16 Solution: Merge sales with employee data

python

```
# Load both datasets
```

```
sales_df = pd.read_csv('sales_data.csv')
```

```
employee_df = pd.read_csv('employee_performance.csv')
```

```
# Merge on employee_id
```

```
merged_df = pd.merge(sales_df, employee_df, on='employee_id', how='left')
```

```
print(merged_df.head())
```

Question 26 Solution: Create customer age groups

python

```
# Load customer data
customer_df = pd.read_csv('customer_data.csv')

# Create age groups
def categorize_age(age):
    if age <= 30:
        return 'Young'
    elif age <= 45:
        return 'Middle'
    else:
        return 'Senior'

customer_df['age_group'] = customer_df['age'].apply(categorize_age)
print(customer_df['age_group'].value_counts())
```

TIPS FOR INTERVIEW SUCCESS

1. Practice data loading with error handling
2. Master groupby operations with multiple aggregations
3. Understand different types of joins and when to use them
4. Be comfortable with datetime operations
5. Know how to handle missing data appropriately
6. Practice explaining your approach and assumptions
7. Understand performance implications of different operations
8. Be prepared to validate your results and catch edge cases

This comprehensive practice set covers all essential pandas skills tested in data analyst interviews. Work through these questions systematically to build confidence and proficiency.